whether constraints were persistently stored [19]. Independent evaluation of *The AI Scientist* further found that 42% of proposed experiments failed to run [32]. Most notably, PaperBench reports that 100% of agent-generated papers contained experimental or methodological weaknesses, and Claude 3.5 Sonnet achieved only 1.8% task completion despite being a frontier model [76].

The following sections formalize these requirements through the auditable autonomous research **AAR** standard (section 4) quantifying when research agents achieve the verifiability necessary for autonomous scientific inquiry [26, 62], and **semantic provenance graphs** (section 5) providing the architectural substrate making auditable autonomous research tractable [11, 48].

4 The AAR standard: A measurement framework for auditable research

In section 3, we discussed three systematic failure modes in autonomous research agents: objective drift (experiments contradict their goals), transient constraints (specifications lost during execution), and unverifiable inference (citations decorrelated from claims). These failures share a common architectural deficit, i.e., the absence of persistent, queryable representations of reasoning that enable systematic verification. Current agents optimize for generation fluency while treating verification as an afterthought, producing systems that are impressively capable yet fundamentally untrustworthy for scientific work [32, 41]. This section establishes the **AAR standard** - a measurement framework defining what properties must be quantified to assess research agent trustworthiness. We argue that auditability must be measurable to be enforceable, and that current benchmarks' exclusive focus on task completion creates perverse incentives that reward unverifiable generation.

4.1 The auditability invariant

We begin with a foundational principle that any research-grade agent must satisfy:

Definition 1 (Research-grade auditability). A research agent is auditable if independent reviewers can verify claim correctness with effort $E_{\text{verify}} \ll E_{\text{generate}}$, where verification accesses only: (1) agent outputs, (2) structured provenance graphs, and (3) cited sources. Systems requiring re-execution of experiments or manual reconstruction of reasoning from logs fail this criterion.

This definition immediately reveals why current systems fail. When independent evaluators identified the AI Scientist's energy efficiency paradox (section 3.1), they spent hours manually tracing through code and logs to reconstruct what the agent actually optimized for [32]. This effort approached that of conducting the original research, defeating the purpose of automation.

4.2 Four properties of auditable research

Each documented failure mode implies a measurable property that, if monitored, would expose problems before outputs propagate to downstream use. We propose four complementary metrics:

Definition 2 (Provenance coverage: Can claims be traced?). The fraction – *PCov* – of claims with complete traceable paths from sources through reasoning steps to outputs [48, 62]. High *PCov* indicates comprehensive traceability; low *PCov* indicates prevalence of orphaned claims lacking verifiable foundations.

Definition 3 (Provenance soundness: Do citations actually support claims?). Semantic validity *PSnd* measuring whether cited sources actually entail attributed claims, not merely whether citations resolve to real papers [21]. *PCov* is necessary but insufficient, i.e., citations may resolve to real papers yet not support the specific relationships asserted, creating scientific pollution: authoritative-looking text with real DOIs but severed logical links between evidence and claim.

Definition 4 (Contradiction transparency: Are evidence conflicts surfaced or suppressed?). The proportion ($CTran$) of actual evidence conflicts that are detected and reported rather than suppressed through aggregation [43, 69]. Vector-based retrieval systems average contradictory embeddings, producing hedged summaries that obscure which specific sources disagree.

4.3 Formal metric definitions

Semantic provenance graph construction. Given a user query q, a deep research agent constructs a directed provenance graph $G_q = (V, E)$ that traces how q is answered through retrieved evidence and reasoning. For each query q, the agent executes a pipeline that retrieves a set of sources ($\mathcal{S}_q$), performs a sequence of reasoning steps ($\mathcal{R}_q$) that synthesize evidence from those sources, and generates a set of final claims ($\mathcal{C}_q$) that directly respond to the query q. The resulting provenance graph is then defined as ($G_q = (V, E)$), where the vertex set is the union of these components, i.e., ($V = \mathcal{S}_q \cup \mathcal{R}_q \cup \mathcal{C}_q$).

Definition 5 (Source nodes). Each $s \in \mathcal{S}_q$ is a tuple (id, DOI/URL, excerpt, timestamp) representing external evidence retrieved for q. Source nodes have no incoming edges.

Definition 6 (Reasoning nodes). Each $r \in \mathcal{R}_q$ is a tuple (id, inference, type, model) where $type \in$ {deduction, induction, synthesis} and *model* identifies the generating LLM.

Definition 7 (Claim nodes). Each $c \in \mathcal{C}_q$ is a tuple (id, statement, location) where *statement* is the factual assertion answering (part of) q and *location* specifies its position in the output. Claim nodes have no outgoing edges.

Definition 8 (Typed edges). Each edge $(u, v, \text{rel}, \nu) \in E$ connects nodes with relation rel $\in$ {supports, contradicts, refines, prerequisite}. For *supports* edges, $\nu \in [0, 1]$ quantifies entailment strength; otherwise ν = null.

Definition 9 (Provenance path). For any claim $c \in \mathcal{C}_q$, the provenance path ($\pi(c)$) is defined as the subgraph obtained by traversing backward from c, where V_c consists of all vertices $v \in V$ that have a directed path to c, and the source and reasoning subsets are given by $S_c = V_c \cap \mathcal{S}_q$ and $R_c = V_c \cap \mathcal{R}_q$, respectively. This path traces how specific sources S_c support claim c through reasoning R_c, establishing the evidential lineage from q to c.

We formalize the four **AAR** properties as computable functions over these sets.

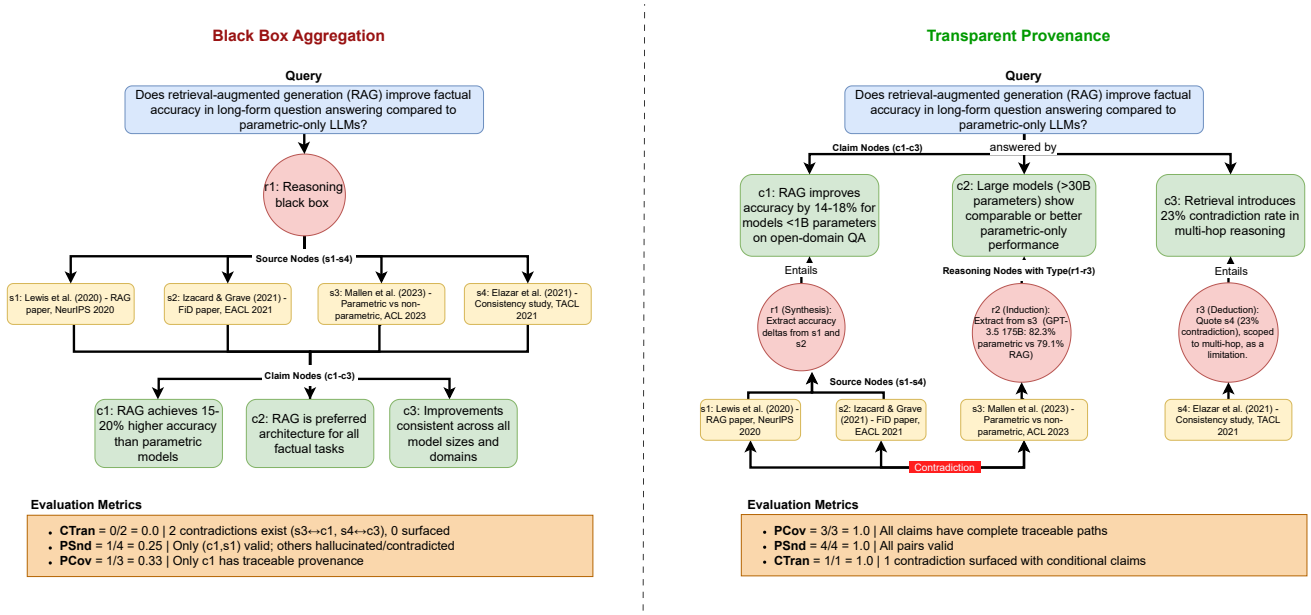

Figure 4: Provenance tracking in RAG Systems. Black-box aggregation (left) versus transparent provenance (right) for identical sources. Black-box reasoning hides two contradictions ($s3 \leftrightarrow c1$, $s4 \leftrightarrow c3$), produces invalid citations, and leaves $c2/c3$ ungrounded – yielding $CTran = 0.0$, $PSnd = 0.25$ (1/4 valid pairs with $\nu > 0.5$), $PCov = 0.33$. Explicit reasoning surfaces model-size contradiction, validates all citations and traces all claims – achieving $CTran = 1.0$, $PSnd = 1.0$, $PCov = 1.0$. Metrics computed using NLI entailment scores (ν).

Provenance coverage: The fraction of claims with complete provenance paths can be calculated as $PCov = \frac{|\{c \in C : \pi(c) \text{ is complete}\}|}{|C|}$

Provenance soundness: For each claim-source pair (c, s) where c cites s, define semantic validity $\nu(c, s) \in [0, 1]$ measuring whether s entails c. Let $\mathcal{P} = \{(c, s) : c \in \mathcal{C}, s \in \mathcal{S}, c \text{ cites } s\}$ be all citation pairs.

$$PSnd = \frac{1}{|\mathcal{P}|} \sum_{(c,s) \in \mathcal{P}} \mathbb{1}[\nu(c, s) > \tau_{\text{entail}}]$$

where τ_{entail} is a domain-specific threshold.

Contradiction transparency: Define $\mathcal{K} = \{(s_i, s_j, e) : s_i, s_j \in \mathcal{S}, e \in \text{entities}\}$ as the set of source pairs making contradictory claims about entity e. Let $\mathcal{D}$ be the subset of contradictions explicitly detected and reported by the system. Quantitatively, $CTran = \frac{|\mathcal{D}|}{|\mathcal{K}|}$.

Audit effort: This is measured empirically through human verification studies. Let T_c be the time (minutes) required for a domain expert to verify claim c given access to: (1) the claim, (2) provenance graph $\pi(c)$, (3) cited sources. The audit effort can be calculated as $AEff = \frac{1}{|C_{\text{sample}}|} \sum_{c \in C_{\text{sample}}} T_c$ where C_{sample} is a representative sample of claims. While $AEff$ measures human time, we can automate this score by simply counting the number of steps and sources in a claim's provenance path. Since the graph is already machine-readable, its structural complexity serves as a fast, automated proxy for how much effort a human would actually need.

Measurement approach: We assess auditability through the above four metrics. First, $PCov$ verifies that claims can be traced to their sources. Second, $PSnd$ uses formal verification to prove whether sources logically entail the attributed claims. Third, $CTran$ identifies contradictions by testing entailment bidirectionally – checking whether evidence supports the claim as well as whether conflicting evidence exists. Fourth, $AEff$ measures verification time empirically through user studies with domain experts [44, 45]. Persistently low **AAR** scores expose concrete architectural deficiencies – low $PCov$ reflects missing provenance infrastructure, low $PSnd$ indicates weak semantic verification, low $CTran$ signals suppressed contradictions, and high $AEff$ denotes poor intelligibility – making these limitations visible and actionable rather than obscured by aggregate performance metrics.

Example. For q = "Does retrieval-augmented generation (RAG) improve factual accuracy in long-form question answering compared to parametric-only LLMs?", the agent retrieves $\mathcal{S}_q = \{s_1, s_2, s_3, s_4\}$ from the literature and generates output claims $\mathcal{C}_q = \{c_1, c_2, c_3\}$ where c_1 states "RAG achieves 14-18% higher accuracy than parametric models", c_2 asserts "RAG is preferred architecture for all factual tasks", and c_3 claims "Improvements generalize across model sizes and domains". Figure 4 contrasts two architectures: *black-box aggregation* (left) directly maps sources to claims without explicit reasoning, achieving $CTran = 0.0$ (contradictions between $s_3 \rightarrow c_1$ and $s_4 \rightarrow c_3$ remain hidden), $PSnd = 0.25$ (only 1 of 4 claim nodes has complete traceable paths), and $PCov = 0.33$ (only c_1 is valid; c_2 and c_3 are hallucinated or contradicted). In contrast, *transparent provenance* (right) exposes intermediate reasoning nodes $\mathcal{R}_q = \{r_1, r_2, r_3\}$ with typed inference (synthesis, induction, deduction), explicitly surfaces the contradiction, and achieves perfect

auditability: $PCov = 1.0$, $PSnd = 1.0$, and $CTran = 1.0$, enabling systematic verification of each claim's evidential support.

5 Perspective: Semantic provenance with protocolized validation

We argue that auditable research agents need provenance graphs with protocolized validation. By this we mean systematic verification that measures entailment strength, source reliability, and extraction confidence according to explicit protocols, operating continuously during synthesis rather than as post-hoc review. Vector-based systems cannot reliably meet $PCov$, $PSnd$, $CTran$, and $AEff$ because embedding similarity does not represent entailment or contradiction. Similar wording can hide opposite claims, and aggregation can erase disagreements. We therefore require provenance that is persistent and queryable. It must record evidence support with measured strength (entailment strength, source reliability, extraction confidence). It must also represent conflicts and show how each conflict was resolved. Finally, it must store reasoning traces across decisions so reviewers can inspect why conclusions were chosen and test counterfactual changes. We show that post-hoc verification does not scale because it cannot recover missing reasoning chains and it arrives too late to prevent error build-up. Validation should run during synthesis with continuous feedback. We also integrate adversarial review into the provenance loop by writing validates/challenges edges that trigger retrieval, revision, or human escalation. Operational gating then blocks unverified claims from entering downstream outputs.

6 Alternative views and objections

The provenance-first architecture faces four recurring objections from practitioners prioritizing immediate deployment over verifiable research infrastructure. We address each systematically.

Bigger models will solve this. Scaling proponents argue that larger models will show emergent capabilities, making external provenance infrastructure unnecessary [33, 70]. We argue this claim conflates generation quality with attribution accuracy. Current systems still exhibit architectural amnesia across both open-weight and proprietary model families [32]. More importantly, provenance is required even when the generated content is correct. If an agent synthesizes conclusions from dozens of papers, users must verify which specific sources support each claim, and this need is independent of model scale. Provenance also enables root-cause analysis when failures occur, supports iterative refinement through queryable audit trails, and provides reproducibility guarantees expected in scientific workflows [30, 62].

Graphs are too expensive. A common objection is that provenance graphs are too expensive to maintain, especially for long-running research workflows, and that flat logs are a more practical option [51]. However, recent evidence suggests that graph-based structure can be cost-effective in practice [10, 11, 15, 31]. Knowledge graph of thoughts reports that graph-based task representations reduce cost while also improving success rates compared to stateless agents [11]. HippoRAG [10] similarly shows that graph-based memory improves multi-hop reasoning that stateless retrieval often misses. Manual post-hoc validation can easily cost more than generation, while architectural validation enables automated graph checks that run in milliseconds per claim. At scale, research agents generate unreliable papers through hallucinated content, and this increases the risk of downstream reuse and training contamination [13, 57]. For this reason, provenance graphs should be viewed as a cost control mechanism where they reduce verification effort, limit error propagation, and lower the long-term cost of maintaining a reliable scientific corpus.

Logs provide sufficient traceability. Some practitioners [7] claim that timestamped logs provide enough traceability because they are easy to collect and have low overhead. We argue that logs only record actions and outputs; they do not capture why an agent makes a choice or how specific evidence supports a claim. For example, when The AI Scientist failed, the logs showed what ran, but they did not explain the key reasoning mistakes, which forced hours of manual inspection to piece together the decision path [32]. In contrast, semantic provenance graphs store explicit links from claims to evidence and reasoning steps, including how the system handles conflicting sources, which enables direct verification queries that flat logs cannot support [30, 62].

Validation introduces prohibitive latency. Critics [4, 34, 65, 72] argue that continuous verification adds latency and compute cost, and they claim this reduces the value of automation. They argue that when verification takes as long as generation, automation loses its efficiency advantage. We disagree because this view treats verification delay as the main cost and ignores larger downstream costs. A small verification delay saves much more time by preventing hallucinated reports that force long debugging and can contaminate later analyses. Independent evaluators of The AI Scientist spend expert effort similar to doing original research to find failures such as the energy efficiency paradox (section 3.1).

7 Conclusion

Deep research agents are becoming capable of generating full scientific narratives at scale, so the limiting factor is no longer writing speed but whether outputs are verifiable before they pollute downstream science. This paper argues that today's logs and citation practices are structurally insufficient because they record actions and references, not the claim-level evidence links needed to audit correctness. We therefore propose "auditable-by-design" research agents built around semantic provenance graphs that explicitly connect sources → intermediate reasoning → final claims in a persistent, queryable form, and we operationalize this goal with the **AAR** standard; provenance coverage (are claims traceable), provenance soundness (do sources actually support claims), contradiction transparency (are conflicts surfaced rather than averaged away), and audit effort (can a human verify faster than redoing the work). Framed this way, progress should be judged not only by task success but by whether verification is cheaper than generation, enabling scalable human oversight and safe autonomy. Looking ahead, the key engineering and research agenda is to standardize provenance schemas, improve entailment/contradiction checking over heterogeneous sources, and build benchmarks that reward low audit cost and high evidence fidelity so that agents produce not just plausible papers, but inspectable scientific artifacts.